

LAYER: InstructPix2Pix Identity-Layer Geometry Results

Stage-B targeted layer optimization with Stage-A identity-layer context

Author: Parth Katiyar

Objective: $Z = 1 - \text{cosine_similarity}(\text{pool}(\text{layer}(\text{original})), \text{pool}(\text{layer}(\text{perturbed})))$. Loss: $\text{loss} = -Z + \text{input_preservation_weight} * \text{MSE}(\text{perturbed}, \text{original})$. Observed preservation weights: 1.

Run matrix

layers	prompts	images	runs	iterations
1.0000	2.0000	4.0000	8.0000	250

Aggregate results

layer	prompt	runs	mean dZ	max dZ	mean SSIM	min SSIM	invalid
unet.conv_in	add black sunglasses	4.0000	-0.0004238	-0.0002488	0.9999	0.9998	0
unet.conv_in	add headphones	4.0000	-0.0004234	-0.0002489	0.9999	0.9998	0

Per-run final values

validity	layer	prompt	image	source	dZ	SSIM	out SSIM	max disp
valid_or_visually_check	unet.conv_in	add black sunglasses	michelle_obama_01	best_valid_checkpoint	-0.0002488	0.9998	0.9995	0.3995
valid_or_visually_check	unet.conv_in	add headphones	michelle_obama_01	best_valid_checkpoint	-0.0002489	0.9998	0.9954	0.3995
valid_or_visually_check	unet.conv_in	add black sunglasses	joe_biden_01	best_valid_checkpoint	-0.0002964	0.9999	0.9996	0.3346
valid_or_visually_check	unet.conv_in	add headphones	joe_biden_01	best_valid_checkpoint	-0.0002964	0.9999	0.9983	0.3346
valid_or_visually_check	unet.conv_in	add headphones	barack_obama_01	best_valid_checkpoint	-0.0005034	0.9998	0.9907	0.3696
valid_or_visually_check	unet.conv_in	add black sunglasses	barack_obama_01	best_valid_checkpoint	-0.0005036	0.9998	0.9994	0.3696
valid_or_visually_check	unet.conv_in	add headphones	kamala_harris_01	best_valid_checkpoint	-0.000645	0.9999	0.9995	0.4585
valid_or_visually_check	unet.conv_in	add black sunglasses	kamala_harris_01	best_valid_checkpoint	-0.0006464	0.9999	0.9997	0.4585

Image strips

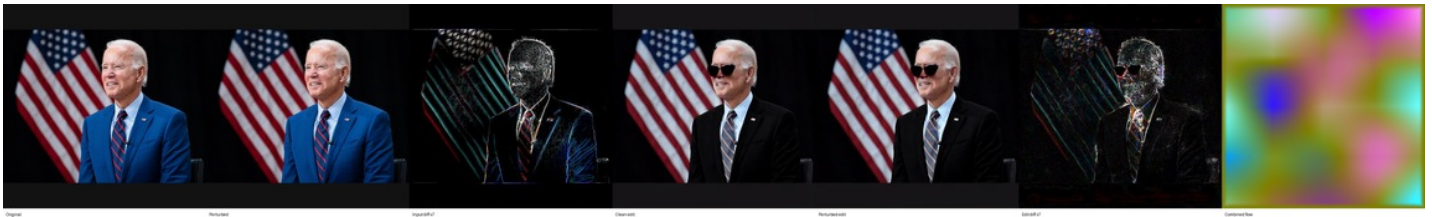
UNET.conv_in / add black sunglasses / michelle_obama_01 - valid_or_visually_check



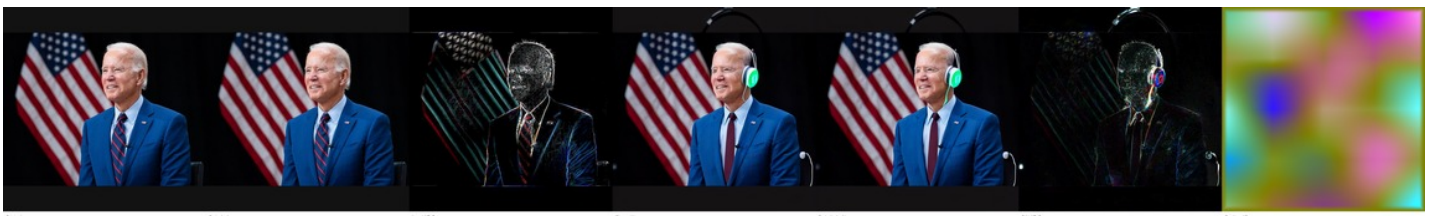
UNET.conv_in / add headphones / michelle_obama_01 - valid_or_visually_check



UNET.conv_in / add black sunglasses / joe_biden_01 - valid_or_visually_check



UNET.conv_in / add headphones / joe_biden_01 - valid_or_visually_check



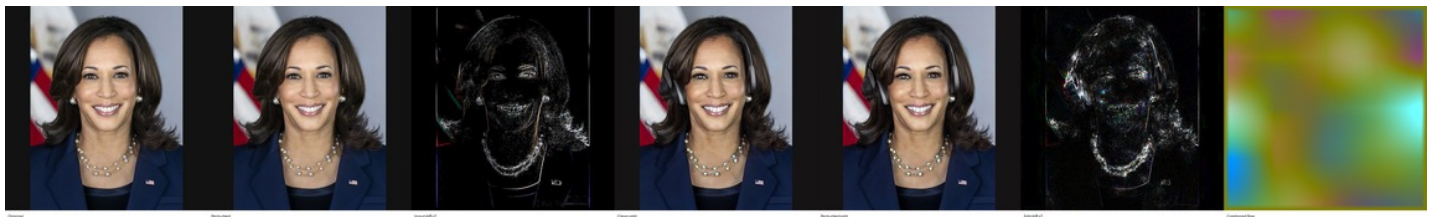
UNET.conv_in / add headphones / barack_obama_01 - valid_or_visually_check



UNET.conv_in / add black sunglasses / barack_obama_01 - valid_or_visually_check



UNET.conv_in / add headphones / kamala_harris_01 - valid_or_visually_check



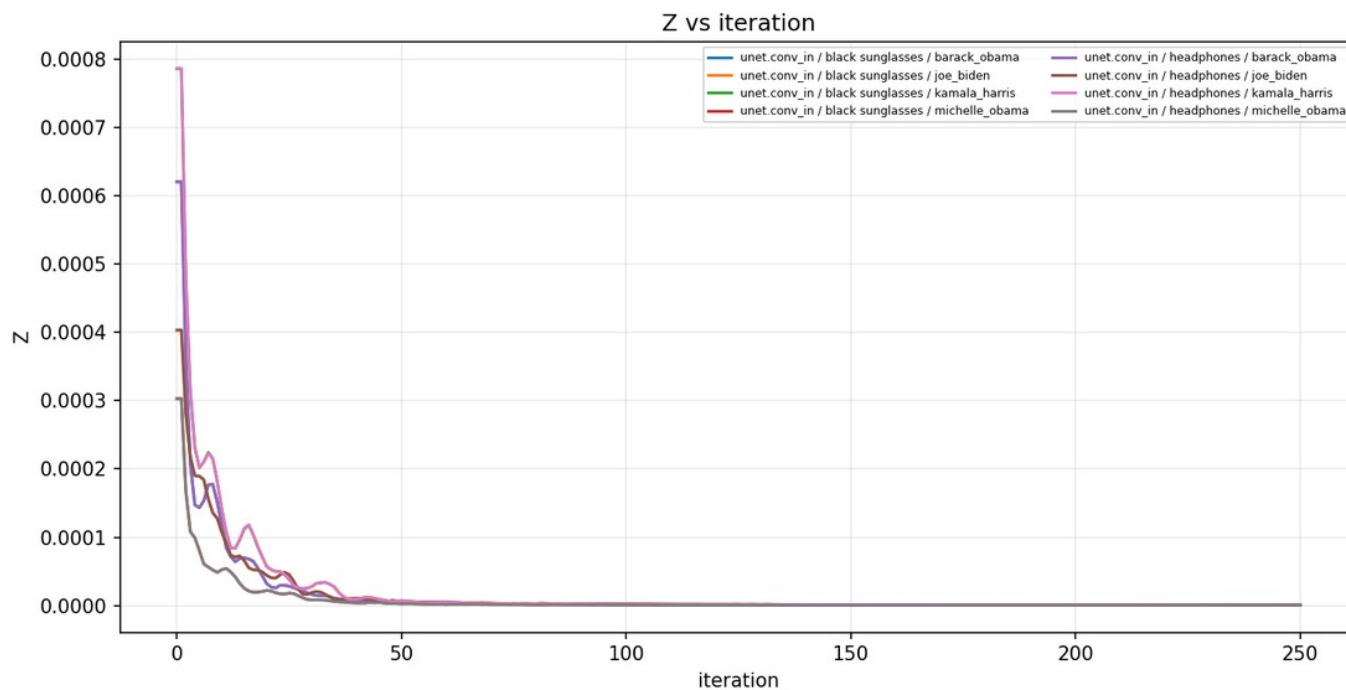
UNET.conv_in / add black sunglasses / kamala_harris_01 - valid_or_visually_check



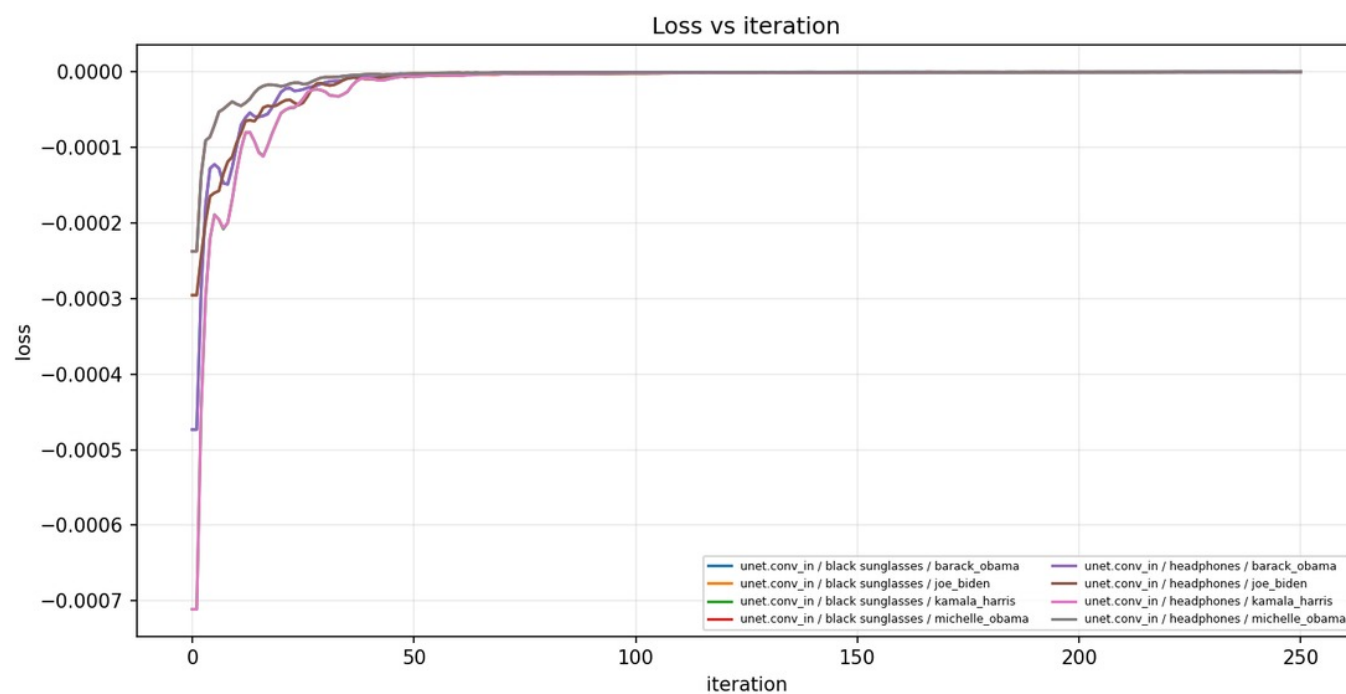
Graphs

Cross-run graphs

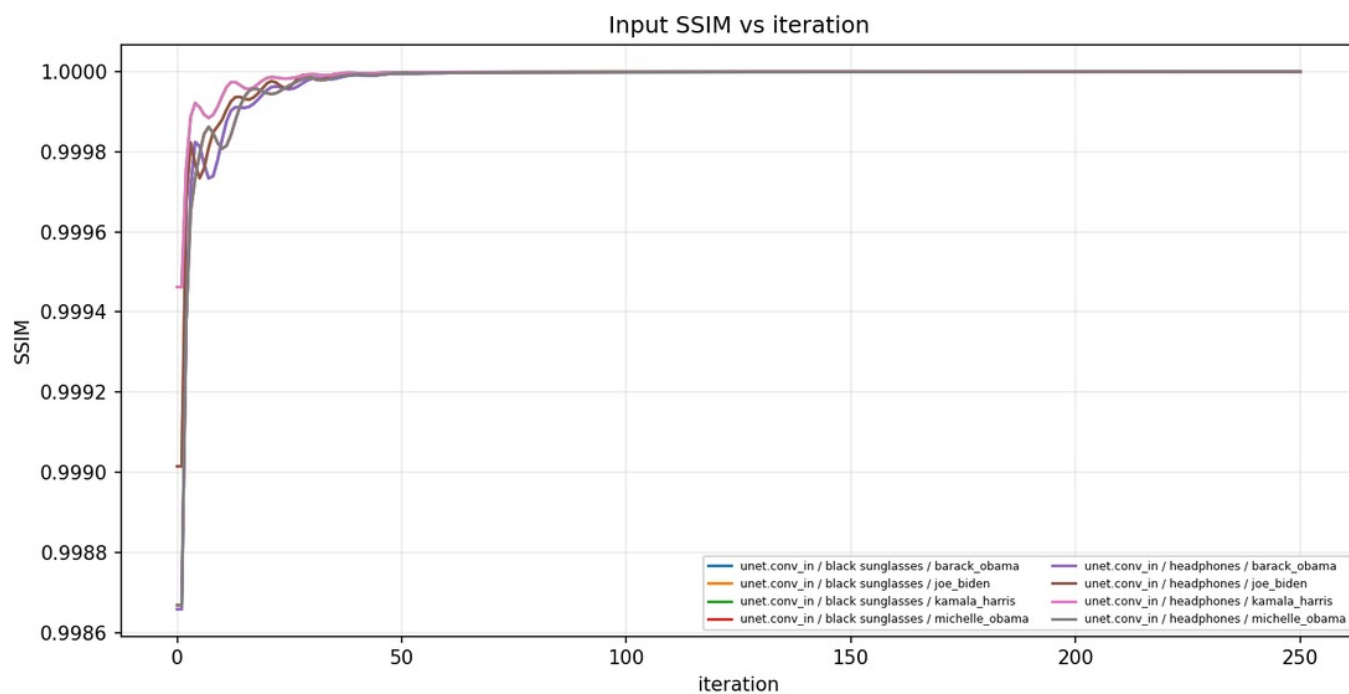
Z vs iteration



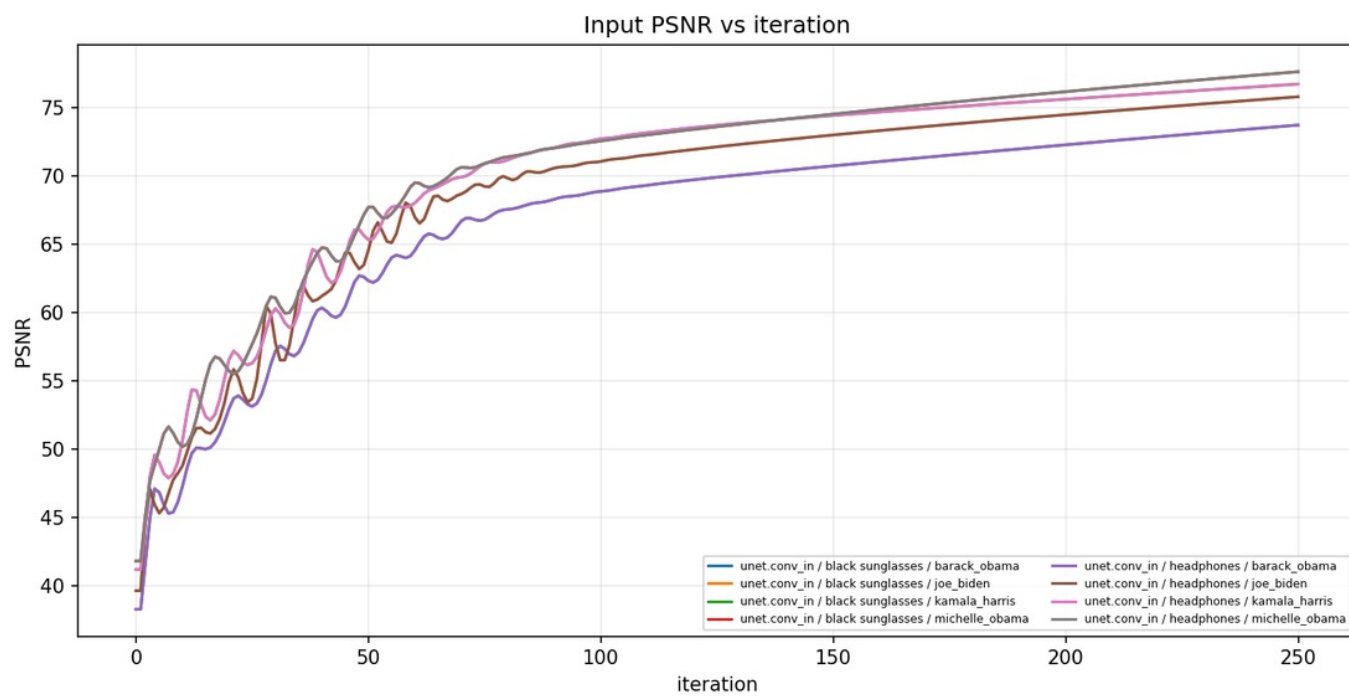
Loss vs iteration



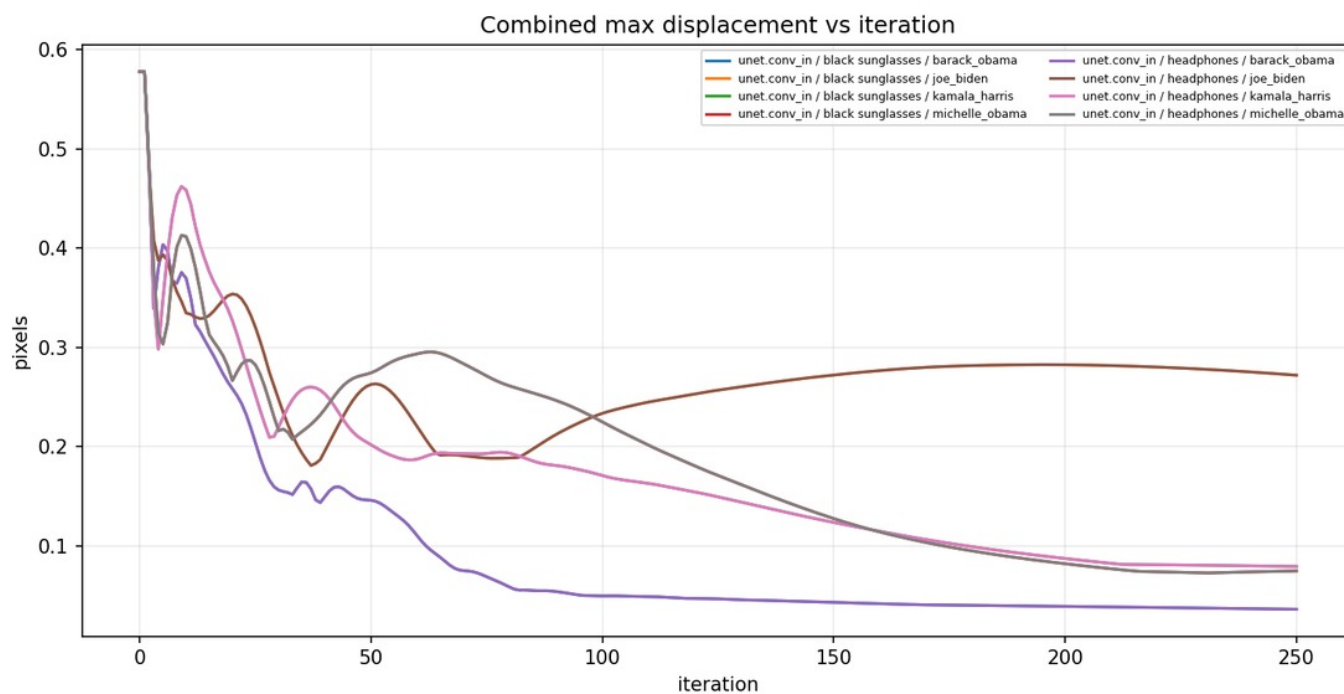
Input SSIM vs iteration



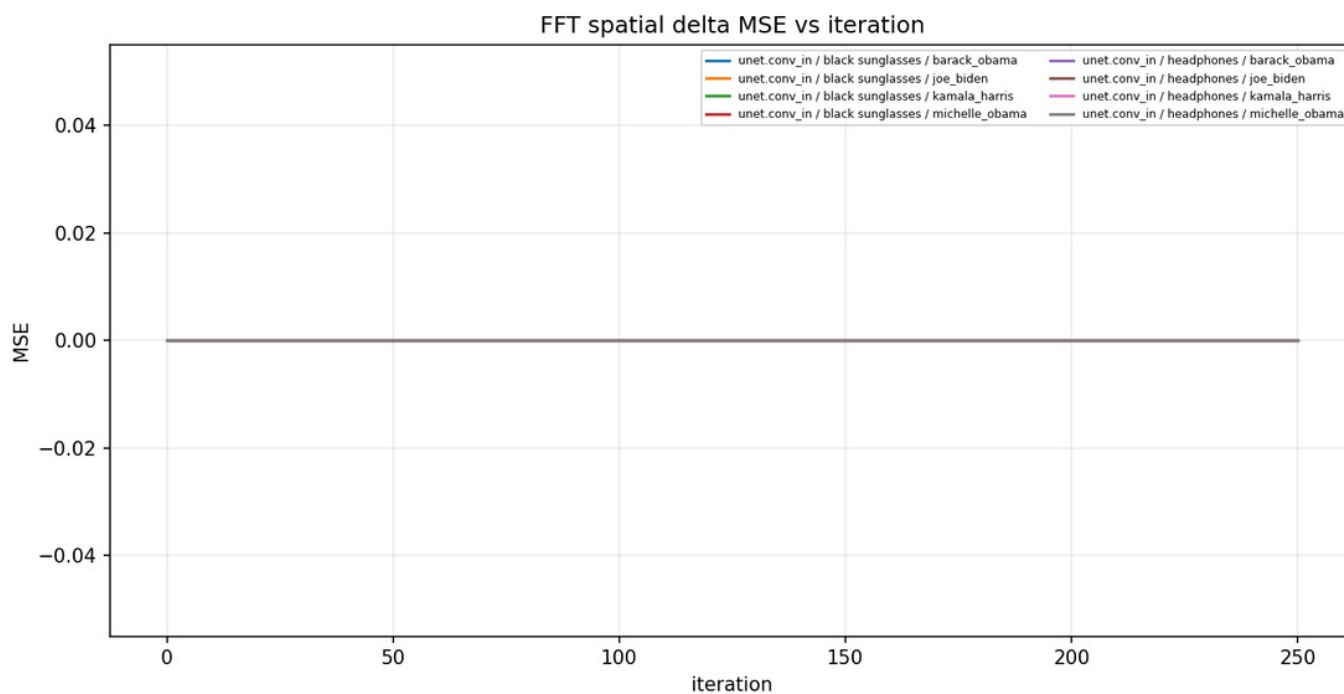
Input PSNR vs iteration



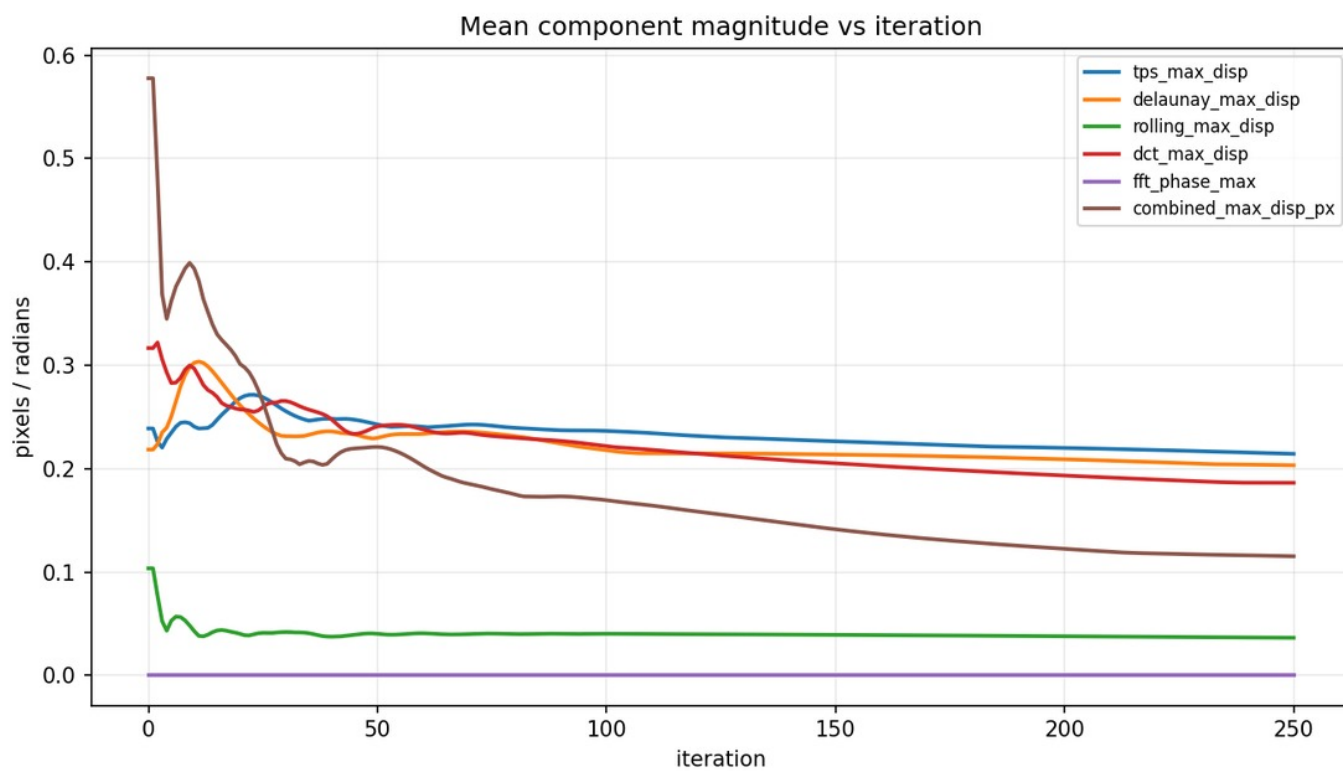
Combined max displacement vs iteration



FFT spatial delta MSE vs iteration



Component magnitude vs iteration



Z increase vs input SSIM

